

VI.

LOAN BANKING

We have so far seen how price levels are determined, showing how they are set by the interaction of the supply of and demand for money. We have seen that the money supply is generally the dominant force in changing prices, while the demand for money is reactive either to long-term conditions or to changes in supply. We have seen, too, that the cause of our chronic inflation is continuing increases in the supply of money, which eventually generate inflationary expectations that aggravate and accelerate the inflation. Eventually, if unchecked, the inflation runs away into a crack-up boom and destruction of the currency. In recent decades, absolute control over the supply of money has been in the hands, not of private enterprise or the free market, but of government.

How does *banking* fit into all this? In what way does banking generate part of the supply of money? Is banking inflationary, and if so, in what sense? How does banking work?

When one speaks of *banks*, there is a semantic problem, since the word bank covers several very different functions and activities. In particular, modern banking mixes and confuses two different operations with very different effects: loans and deposits. Let

us first see how what we might call *loan banking* originated and what its relationship might be to the money supply and to inflation.

Most people *think* of banks as institutions which channel their savings into productive loans and investments. *Loan banking* is essentially that healthy and productive process in operation.

Let's see how it works. Suppose that I have saved \$10,000 and have decided to set up a loan business, or what we might call a loan bank.¹ I set up the Rothbard Loan Company.

A *must* in making any sense whatever out of the banking system is to become familiar with the common accounting device of the *T-account*, or balance sheet. The balance sheet is a product of one of the most important inventions of modern civilization: double-entry bookkeeping, which came to Renaissance Italy from the Arab civilization of North Africa. Before double-entry bookkeeping, business firms kept single-entry books, which were simply running accounts of expenditures, income, and so on. They found it impossible to know where they had made mistakes, and therefore could not try to correct them. Double-entry bookkeeping, on the other hand, often means that any entry on one side of the ledger must immediately, and automatically, be balanced by an entry on the other side, the totals of which must be identical. It then becomes relatively easy to find out where the totals do not balance, and therefore where the error has occurred.

While the concept of double-entry bookkeeping was established during the Renaissance, the familiar T-account balance sheet was formalized only at the start of the "classical" period of modern accounting, that is, the late nineteenth century.²

¹We are using "dollars" instead of "gold ounces," because this process is the same whether we are on a gold or a fiat standard.

²In particular, the originator of the Assets = Liability + Equity equation was the distinguished American accountant, Charles E. Sprague, who conceived the idea in 1880 and continued to advance the idea until after the turn of the century. See Gary J. Previts and Barbara D. Merino, *A History of Accounting in America* (New York: Ronald Press, 1979), pp. 107–13.

On the T-account balance sheet, the left side is the monetary valuation, at any given time, of the total assets of the business firm. This side is, appropriately enough, labeled “Assets.” On the right side we have the total amount of assets owned by one or more owners. In short, any and all assets must be owned by *some-one*, so that if we add up the assets owned by A, B, C . . . etc., they should yield a total identical to the total sum of the assets. Some assets are owned in fact by the owner or owners of the firm (Equity Capital). Others are owed to, and therefore in an economic sense claimed or owned by, various creditors of the firm (Liabilities). So that, as total assets are apportioned among the various owners or claimants, the total of the right column, “Equity plus Liabilities,” must precisely equal the total assets on the left side.

Let us now return to the Rothbard Loan Company. I have saved \$10,000 in cash, and place it in my firm’s account. The balance sheet of the new company is now as follows:

Assets		Equity & Liabilities	
Cash	\$10,000	<i>Equity</i>	
		Rothbard	\$10,000
Total	\$10,000	Total	\$10,000

FIGURE 6.1 — STARTING THE LOAN BANK

The T-account shows that the assets of the Rothbard Loan Company are now \$10,000 in cash, and that I own these assets. Total assets are precisely equal to total assets owned.

The purpose of forming the Rothbard Loan Company is, of course, to lend money out and to earn interest. Suppose that I now lend \$9,000 to Joe’s Diner for a new counter, keeping

\$1,000 as a cash reserve. Joe borrows \$9,000 at 10 percent interest, promising to pay me back \$9,900 in one year's time. In short, I give Joe \$9,000, in return for which he gives me an IOU for \$9,900 for one year in the future. My asset is now an IOU from Joe to be realized in the future. The balance sheet of the Rothbard Loan Company is now as follows:

Assets		Equity & Liabilities	
Cash	\$1,000		
IOU from Joe	9,900		
		<i>Equity</i>	
		Rothbard	\$10,900
Total	\$10,900	Total	\$10,900

FIGURE 6.2 — MAKING A LOAN

My assets have now happily grown, at least in anticipation. Total assets and equity are now \$10,900. What, in all of this, has happened to the total supply of money so far? The answer is, nothing. Let us say that there was at the onset of the Rothbard Loan Company, \$10,000 in circulation. I saved \$10,000, and then loaned \$9,000 to Joe. The money supply has in no sense increased; some of mine has simply been saved (that is, not spent on consumer goods), and loaned to someone who will spend it, in this case on productive investment.

Let us now see what happens one year later when Joe repays the \$9,900. The IOU is canceled, and I now have in cash the loan paid back plus interest (Figure 6.3).

The loan is repaid, and my firm, and therefore myself, is \$900 richer. But, once again, there has been no increase in society's stock of money. For in order to pay back the loan, Joe had to save \$900 out of profits. Again, Joe and I are transferring to each

Assets		Equity & Liabilities	
Cash	\$10,900	<i>Equity</i>	
		Rothbard	\$10,900
Total	\$10,900	Total	\$10,900

FIGURE 6.3 — THE LOAN AS PAID

other the ownership of existing cash balances which we have saved by not consuming. My loan bank has channeled savings into loans, the loans have been repaid, and at no point has the money supply increased. Loan banking is a productive, noninflationary institution.

The loan to Joe did not have to be made for business investment. It could have been a loan for consumption purposes, say, to enable him to buy a new car. Joe anticipates having higher income or lower expenditures next year, enabling him to pay back the loan with interest. In this case, he is not so much making a monetary profit from the loan as rearranging the time pattern of his expenditures, paying a premium for the use of money now rather than having to wait to buy the car. Once again, the total money supply has not changed; money is being saved by me and my firm, and loaned to Joe, who then saves enough of the existing money supply to fulfill his contractual obligations. Credit, and loan banking, is productive, benefits both the saver and the borrower, and causes no inflationary increase in the money supply.

Suppose now that my loan bank is flourishing and I expand the firm by taking in a partner, my brother-in-law, who contributes another \$10,900 in cash to the firm. The Rothbard Loan Bank now looks as follows:

Assets		Equity & Liabilities	
Cash	\$21,800	<i>Equity</i>	
		Rothbard	\$10,900
		Brother-in-law	10,900
Total	\$21,800	Total	\$21,800

FIGURE 6.4 — EXPANDING BANK EQUITY

The firm has now expanded, and the increased assets are owned equally by my brother-in-law and me. Total assets, and total assets owned, have grown equally and accordingly. Once again, there has been no increase in the stock of money, for my brother-in-law has simply saved \$10,900 from the existing supply, and invested it. Then, when more loans are made, cash shifts into IOUs and interest receipts eventually add to cash, total assets, and equity.

As the loan bank expands, we might decide to keep raising capital by expanding the number of partners, or perhaps by converting to a joint-stock company (legally, a corporation), which issues low-denomination stock and can thereby tap the savings of small investors. Thus, we might set up the Rothbard Loan Bank Corporation, which sells 10,000 shares at \$10 apiece, and thereby accumulates \$100,000 for making loans. Assume that \$95,000 is loaned out and \$5,000 kept in cash. The balance sheet of the Rothbard Loan Bank Corporation would now be as shown in Figure 6.5.

We could list the shareholders, and how many shares thus owned in proportion to the total assets of the newly-expanded Rothbard Loan Corporation. We won't, because the important point is that more savings have been channeled into productive credit, to earn an interest return. Note that there has been no

increase in the supply of money, and therefore no impetus toward inflation.

Assets		Equity & Liabilities	
Cash	\$ 5,000		
IOUs	\$ 95,000		
		<i>Equity</i>	
		Shareholders	\$100,000
Total	\$100,000	Total	\$21,800

FIGURE 6.5 — BANK GOING PUBLIC

Let us now expand the bank further. In addition to shareholders, the Rothbard Bank now decides to float bonds or other debentures, and thereby borrow from some people in order to lend to others. Let us assume that the Rothbard Bank issues \$50,000 worth of bonds, and sells them on the bond market. The bonds are to be repaid in 20 years, paying 10 percent per year on their face value. Now \$50,000 in cash is added to the bank's coffers. We can also sell certificates of deposit, a relatively new banking instrument in which the owner of the certificate, Jones, buys a certificate worth \$20,000 for six months, at 10 percent interest. In effect, Jones lends the Rothbard bank \$20,000 in exchange for the bank's IOU that it will repay Jones \$21,000 in six months' time. The Rothbard Bank borrows these moneys because it expects to be able to lend the new cash at a greater than 10 percent rate, thus earning a profit differential between the interest it pays out and the interest it earns. Suppose it is able to lend the new money at 15 percent interest, thereby making a profit of 5 percent on these transactions. If its administrative expenses of operation are, say, 2 percent, it is able to make a 3 percent profit on the entire transaction.

The new balance sheet of the Rothbard Bank, after it has issued \$50,000 worth of long-term bonds, and sold a \$20,000 short-term certificate of deposit to Jones, looks like this:

Assets		Equity & Liabilities	
		<i>Liabilities</i>	
Cash	\$75,000	Bonds	\$50,000
IOUs	\$95,000	Certificate of deposit to Jones	\$20,000
		Total	\$70,000
		<i>Equity</i>	
		Shareholders	\$100,000
Total	\$170,000	Total	\$170,000

FIGURE 6.6 — BANK ISSUING DEBENTURES

The balance sheet of the Rothbard Bank has now become far more complex. The assets, cash and IOUs are owned or claimed by a combination of people: by the legal owners, or equity, and by those who have money claims on the bank. In the economic sense, the legal owners and the creditors jointly *own* part of the Rothbard Bank, because they have joint claims on the bank's assets. To the shareholders' invested \$100,000 are now added \$50,000 borrowed from bondholders and a \$20,000 CD (certificate of deposit) sold to Jones. Once again, of course, the Rothbard Bank takes the newly acquired cash and lends it for further IOUs, so that the balance sheet now looks like Figure 6.7.

The Rothbard Bank is now doing exactly what most people think banks always do: borrowing money from some (in addition to investing the savings of the owners) and lending money to others. The bank makes money on the interest differential because it

Assets		Equity & Liabilities	
		<i>Liabilities</i>	
Cash	\$5,000	Bonds	\$50,000
IOUs	\$165,000	CD	\$20,000
		Total	\$70,000
		<i>Equity</i>	
		Shareholders	\$100,000
Total	\$170,000	Total	\$170,000

FIGURE 6.7 — BANK LENDING BORROWED FUNDS

is performing the important social service of channeling the borrowed savings of many people into productive loans and investments. The bank is expert on where its loans should be made and to whom, and reaps the reward for this service.

Note that there has still been no inflationary action by the loan bank. No matter how large it grows, it is still only tapping savings from the existing money stock and lending that money to others.

If the bank makes unsound loans and goes bankrupt, then, as in *any* kind of insolvency, its shareholders and creditors will suffer losses. This sort of bankruptcy is little different from any other: unwise management or poor entrepreneurship will have caused harm to owners and creditors.

Factors, investment banks, finance companies, and money-lenders are just some of the institutions that have engaged in loan banking. In the ancient world, and in medieval and pre-modern Europe, most of these institutions were forms of “moneylending proper,” in which owners loaned out their own saved money. Loan banks, in the sense of intermediaries, borrowing from savers to lend to borrowers, began only in Venice in the late Middle Ages. In England, intermediary-banking began only with the

“scriveners” of the early seventeenth century.³ The scriveners were clerks who wrote contracts and bonds, and were therefore often in a position to learn of mercantile transactions and engage in moneylending and borrowing. By the beginning of the eighteenth century, scriveners had been replaced by more advanced forms of banking.

³During the sixteenth century, most English moneylending was conducted, not by specialized firms, but by wealthy merchants in the clothing and woolen industries, as an outlet for their surplus capital. See J. Milnes Holden, *The History of Negotiable Instruments in English Law* (London: The Athlone Press, 1955), pp. 205–06.

VII.

DEPOSIT BANKING

1. WAREHOUSE RECEIPTS

Deposit banking began as a totally different institution from loan banking. Hence it was unfortunate that the same name, *bank*, became attached to both. If loan banking was a way of channeling savings into productive loans to earn interest, deposit banking arose to serve the convenience of the holders of gold and silver. Owners of gold bullion did not wish to keep it at home or office and suffer the risk of theft; far better to store the gold in a safe place. Similarly, holders of gold coin found the metal often heavy and inconvenient to carry, and needed a place for safekeeping. These deposit banks functioned very much as safe-deposit boxes do today: as safe “money warehouses.” As in the case of any warehouse, the *depositor* placed his goods *on deposit* or in trust at the warehouse, and in return received a ticket (or warehouse receipt) stating that he could redeem his goods whenever he presented the ticket at the warehouse. In short, his ticket or receipt or claim check was to be instantly redeemable *on demand* at the warehouse.

Money in a warehouse can be distinguished from other deposited goods, such as wheat, furniture, jewelry, or whatever. All of these goods are likely to be redeemed fairly soon after storage, and then revert to their regular use as a consumer or capital good. But gold, apart from jewelry or industrial use, largely serves as money, that is, it is only *exchanged* rather than used in consumption or production. Originally, in order to use his gold for exchange, the depositor would have to redeem his deposit and then turn the gold over to someone else in exchange for a good or service. But over the decades, one or more money warehouses, or deposit banks, gained a reputation for probity and honesty. Their warehouse receipts then began to be transferred directly as a surrogate for the gold coin itself. The warehouse receipts were scrip for the real thing, in which metal they could be redeemed. They functioned as “gold certificates.”¹ In this situation, note that the total money supply in the economy has not changed; only its *form* has altered. Suppose, for example, that the initial money supply in a country, when money is only gold, is \$100 million. Suppose now that \$80 million in gold is deposited in deposit banks, and the warehouse receipts are now used as proxies, as substitutes, for gold. In the meanwhile, \$20 million in gold coin and bullion are left outside the banks in circulation. In this case, the total money supply is still \$100 million, except that now the money in circulation consists of \$20 million in gold coin and \$80 million in gold certificates standing in for the actual \$80 million of gold in bank vaults. Deposit banking, when the banks really act as genuine money warehouses, is still eminently productive and noninflationary.

How can deposit banks charge for this important service? In the same way as any warehouse or safe-deposit box: by charging a fee in proportion to the time that the deposit remains in the

¹Dobie writes: “a transfer of the warehouse receipt, in general confers the same measure of title that an actual delivery of the goods which it represents would confer.” Armistead M. Dobie, *Handbook on the Law of Bailments and Carriers* (St. Paul, Minn.: West Publishing Co., 1914), p. 163.

bank vaults. There should be no mystery or puzzlement about *this* part of the banking process.

How do these warehouse receipt transactions relate to the T-account balance sheets of the deposit banks? In simple justice, not at all. When I store a piece of furniture worth \$5,000 in a warehouse, in law and in justice the furniture does *not* show up as an asset of the warehouse during the time that I keep it there.

The warehouse does not add \$5,000 to both its assets and liabilities because it in no sense *owns* the furniture; neither can we say that I have *loaned* the warehouse the furniture for some indefinite time period. The furniture is mine and remains mine; I am only keeping it there for safekeeping and therefore I am legally and morally entitled to redeem it any time I please. I am not therefore the bank's "creditor"; it doesn't *owe* me money which I may some day collect. Hence, there is no debt to show up on the Equity + Liability side of the ledger. Legally, the entire transaction is not a loan but a *bailment*, hiring someone for the safekeeping of valuables.

Let us see why we are dealing with a bailment, not a loan. In a loan, or a *credit transaction*, the creditor exchanges a *present good*—that is, a good available for use at any time in the present—for a *future good*, an IOU redeemable at some date in the future. Since present goods are more valuable than future goods, the creditor will invariably charge, and the debtor pay, an interest premium for the loan.

The hallmark of a loan, then, is that the money is due at some future date and that the debtor pays the creditor interest. But the deposit, or *claim transaction*, is precisely the opposite. The money must be paid by the bank *at any time* the depositor presents the ticket, and not at some precise date in the future. And the bank—the alleged "borrower" of the money—generally does not pay the depositor for making the loan. Often, it is the depositor who pays the bank for the service of safeguarding his valuables.

Deposit banking, or money warehousing, was known in ancient Greece and Egypt, and appeared in Damascus in the early thirteenth century, and in Venice a century later. It was prominent

in Amsterdam and Hamburg in the seventeenth and eighteenth centuries.

In England, there were no banks of deposit until the Civil War in the mid-seventeenth century. Merchants were in the habit of keeping their surplus gold in the king's mint in the Tower of London—an institution which of course was accustomed to storing gold. The habit proved to be an unfortunate one, for when Charles I needed money in 1638 shortly before the outbreak of the Civil War, he simply confiscated a large sum of gold, amounting to £200,000, calling it a “loan” from the depositors. Although the merchants finally got their money back, they were understandably shaken by the experience, and forsook the mint, instead depositing their gold in the coffers of private goldsmiths, who were also accustomed to the storing and safekeeping of the valuable metal.² The goldsmith's warehouse receipts then came to be used as a surrogate for the gold money itself.³

All men are subject to the temptation to commit theft or fraud, and the warehousing profession is no exception. In warehousing, one form of this temptation is to steal the stored products outright—to skip the country, so to speak, with the stored gold and jewels. Short of this thievery, the warehouse man is subject to a more subtle form of the same temptation: to steal or “borrow” the valuables “temporarily” and to profit by speculation or whatever, returning the valuables before they are redeemed so that no one will be the wiser. This form of theft is known as *embezzlement*,

²The business of the goldsmiths was to manufacture gold and silver plate and jewelry, and to purchase, mount and sell jewels. See J. Milnes Holden, *The History of Negotiable Instruments in English Law* (London: The Athlone Press, 1955), pp. 70–71.

³These were two other reasons for the emergence of the goldsmiths as money warehouses during the Civil War. Apprentices, who had previously been entrusted with merchants' cash, were now running off to the army, so that merchants now turned to the goldsmiths. At the same time, the gold plate business had fallen off, for impoverished aristocrats were melting down their gold plate for ready cash instead of buying new products. Hence, the goldsmiths were happy to turn to this new form of business. *Ibid.*

which the dictionary defines as “appropriating fraudulently to one’s own use, as money or property entrusted to one’s care.”

But the speculating warehouseman is always in trouble, for the depositor can come and present his claim check *at any time*, and he is legally bound to redeem the claim, to return the valuables instantly on demand. Ordinarily, then, the warehousing business provides little or no room for this subtle form of theft. If I deposit a gold watch or a chair in a warehouse, I want the object when I call for it, and if it isn’t there, the warehouseman will be on a trip to the local prison.

In some forms of warehousing, the temptation to embezzle is particularly heady. The depositor is here not so much interested in getting back the *specific* object as he is in receiving the *same kind* of product. This will occur in the case of *fungible* commodities such as grain, where each unit of the product is identical to every other. Such a deposit is a “general” rather than a “specific” deposit warrant. It now becomes more convenient for the warehouseman to mix all bushels of grain of the same type into a common bin, so that anyone redeeming his grain receives bushels from the same bin. But now the temptation to embezzle has increased enormously. All the warehouseman need do is arrive at a workable estimate of what *percentage* of the grain will probably be redeemed in the next month or year, and then he can lend out or speculate on the rest.

In sophisticated transactions, however, the warehouseman is not likely physically to remove the grain. Since warehouse receipts serve as surrogates for the grain itself, the warehouseman will instead print fake, or *counterfeit*, warehouse receipts, which will look exactly like the others.

But, it might be asked, what about the severe legal penalties for embezzlement? Isn’t the threat of criminal charges and a jail term enough to deter all but the most dedicated warehouse embezzlers? Perhaps, except for the critical fact that bailment law scarcely existed until the eighteenth century. It was only by the twentieth century that the courts finally decided that the grain warehouseman was truly a bailee and not simply a debtor.

2. DEPOSIT BANKING AND EMBEZZLEMENT

Gold coin and bullion—money—provides an even greater temptation for embezzlement to the deposit banker than grain to the warehouseman. Gold coin and bullion are fully as fungible as wheat; the gold depositor, too, unless he is a collector or numismatist, doesn't care about receiving the identical gold coins he once deposited, so long as they are of the same mark and weight. But the temptation is even greater in the case of money, for while people *do* use up wheat from time to time, and transform it into flour and bread, gold as money *does not have to be used at all*. It is only employed in exchange and, so long as the bank continues its reputation for integrity, its warehouse receipts can function very well as a surrogate for gold itself. So that if there are few banks in the society and banks maintain a high reputation for integrity, there need be little redemption at all. The confident banker can then estimate that a smaller part of his receipts will be redeemed next year, say 15 percent, while fake warehouse receipts for the other 85 percent can be printed and loaned out without much fear of discovery or retribution.

The English goldsmiths discovered and fell prey to this temptation in a very short time, in fact by the end of the Civil War. So eager were they to make profits in this basically fraudulent enterprise, that they even offered to pay interest to depositors so that they could then “lend out” the money. The “lending out,” however, was duplicitous, since the depositors, possessing their warehouse receipts, were under the impression that their money was safe in the goldsmiths' vaults, and so exchanged them as equivalent to gold. Thus, gold in the goldsmiths' vaults was covered by two or more receipts. A genuine receipt originated in an actual deposit of gold stored in the vaults, while counterfeit ones, masquerading as genuine receipts, had been printed and loaned out by goldsmiths and were now floating around the country as surrogates for the same ounces of gold.⁴

⁴See *ibid.*, p. 72.

The same process of defrauding took place in one of the earliest instances of deposit banking: ancient China. Deposit banking began in the eighth century, when shops accepted valuables and received a fee for safekeeping. After a while, the deposit receipts of these shops began to circulate as money. Finally, after two centuries, the shops began to issue and hand out more printed receipts than they had on deposit; they had caught onto the deposit banking scam.⁵ Venice, from the fourteenth to the sixteenth centuries, struggled with the same kind of bank fraud.

Why, then, were the banks and goldsmiths not cracked down on as defrauders and embezzlers? Because deposit banking law was in even worse shape than overall warehouse law and moved in the opposite direction to declare money deposits not a bailment but a debt.

Thus, in England, the goldsmiths, and the deposit banks which developed subsequently, boldly printed counterfeit warehouse receipts, confident that the law would not deal harshly with them. Oddly enough, no one tested the matter in the courts during the late seventeenth or eighteenth centuries. The first fateful case was decided in 1811, in *Carr v. Carr*. The court had to decide whether the term “debts” mentioned in a will included a cash balance in a bank deposit account. Unfortunately, Master of the Rolls Sir William Grant ruled that it did. Grant maintained that since the money had been paid generally into the bank, and was not earmarked in a sealed bag, it had become a loan rather than

⁵By A.D. 700–800 there were shops in China which would accept valuables and, for a fee, keep them safe. They would honour drafts drawn on the items in deposit, and, as with the goldsmith’s shops in Europe, their deposit receipts gradually began to circulate as money. It is not known how rapidly this process developed, but by A.D. 1000 there were apparently a number of firms in China which issued regular printed notes and which had discovered that they could circulate more notes than the amount of valuables they had on deposit.

Tullock, “Paper Money: A Cycle in Cathay,” *Economic History Review* 9 (August 1957): 396.

a bailment.⁶ Five years later, in the key follow-up case of *Devaynes v. Noble*, one of the counsel argued, correctly, that “a banker is rather a bailee of his customer’s funds than his debtor . . . because the money in . . . [his] hands is rather a deposit than a debt, and may therefore be instantly demanded and taken up.” But the same Judge Grant again insisted—in contrast to what would be happening later in grain warehouse law—that “money paid into a banker’s becomes immediately a part of his general assets; and he is merely a debtor for the amount.”⁷

The classic case occurred in 1848 in the House of Lords, in *Foley v. Hill and Others*. Asserting that the bank customer is only its creditor, “with a superadded obligation arising out of the custom (sic?) of the bankers to honour the customer’s cheques,” Lord Cottenham made his decision, lucidly if incorrectly and even disastrously:

Money, when paid into a bank, ceases altogether to be the money of the principal; it is then the money of the banker, who is bound to an equivalent by paying a similar sum to that deposited with him when he is asked for it. . . . The money placed in the custody of a banker is, to all intents and purposes, the money of the banker, to do with it as he pleases; he is guilty of no breach of trust in employing it; he is not answerable to the principal if he puts it into jeopardy, if he engages in a hazardous speculation; he is not bound to keep it or deal with it as the property of his principal; but he is, of course, answerable for the amount, because he has contracted.⁸

Thus, the banks, in this astonishing decision, were given *carte blanche*. Despite the fact that the money, as Lord Cottenham

⁶*Carr v. Carr* (1811) 1 Mer. 543. In J. Milnes Holden, *The Law and Practice of Banking*, vol. I, *Banker and Customer* (London: Pitman Publishing, 1970), p. 31.

⁷*Devaynes v. Noble* (1816) 1 Met. 529; in *ibid.*

⁸*Foley v. Hill and Others* (1848) 2. H.L.C., pp. 36–37; in *ibid.*, p. 32.

conceded, was “placed in the custody of the banker,” he can do virtually anything with it, and if he cannot meet his contractual obligations he is only a legitimate insolvent instead of an embezzler and a thief who has been caught red-handed. To *Foley* and the previous decisions must be ascribed the major share of the blame for our fraudulent system of *fractional reserve banking* and for the disastrous inflations of the past two centuries.

Even though American banking law has been built squarely on the *Foley* concept, there are intriguing anomalies and inconsistencies. While the courts have insisted that the bank deposit is only a debt contract, they still try to meld in something more. And the courts remain in a state of confusion about whether or not a deposit—the “placing of money in a bank for safekeeping”—constitutes an *investment* (the “placing of money in some form of property for income or profit”). For if it is purely safekeeping and not investment, then the courts might one day be forced to concede, after all, that a bank deposit is a bailment; but if an investment, then how do safekeeping and redemption on demand fit into the picture?⁹

Furthermore, if only *special bank* deposits where the identical object must be returned (e.g., in one’s safe-deposit box) are to be considered bailments, and general bank deposits are debt, then why doesn’t the same reasoning apply to other fungible, general deposits such as wheat? Why aren’t wheat warehouse receipts

⁹See *Michie on Banks and Banking*, rev. ed. (Charlottesville, Va.: Michie Co., 1973), vol. 5A, p. 20. Also see pp. 1–13, 27–31, and *ibid.*, 1979 *Cumulative Supplement*, pp. 3–4, 7–9. Thus, Michie states that a “bank deposit is more than an ordinary debt, and the depositor’s relation to the bank is not identical with that of an ordinary creditor.” Citing a Pennsylvania case, Michie adds that “a bank deposit is different from an ordinary debt in this, that from its very nature it is constantly subject to the check of the depositor, and is always payable on demand”. *People’s Bank v. Legrand*, 103 Penn.309, 49 Am.R.126. Michie, *Banks and Banking*, p. 13n. Also, despite the laws insistence that a bank “becomes the absolute owner of money deposited with it,” a bank still “cannot speculate with its depositors’ money.” *Banks and Banking*, pp. 28, 30–31.

only a debt? Why is this inconsistent law, as the law concedes, “peculiar to the banking business”?^{10,11}

3. FRACTIONAL RESERVE BANKING

The *carte blanche* for deposit banks to issue counterfeit warehouse receipts for gold had many fateful consequences. In the first place, it meant that any deposit of money *could* now take its place in the balance sheet of the bank. For the duration of the deposit, the gold or silver now became an owned asset of the bank, with redemption due as a supposed *debt*, albeit instantly on demand. Let us assume we now have a Rothbard Deposit Bank. It opens for business and receives a deposit of \$50,000 of gold from Jones, for which Jones receives a warehouse receipt which he may redeem on demand at any time. The balance sheet of the Rothbard Deposit Bank is now as shown in Figure 7.1.

Although the first step has begun on the slippery slope to fraudulent and deeply inflationary banking, the Rothbard Bank has not yet committed fraud or generated inflation. Apart from a general deposit now being considered a debt rather than bailment,

¹⁰Michie, *Banks and Banking*, p. 20. The answer of the distinguished legal historian Arthur Nussbaum is that the “contrary view” (that a bank deposit is a bailment not a debt) “would lay an unbearable burden upon banking business.” No doubt exuberant bank profits from issue of fraudulent warehouse receipts would come to an end. But grain elevators and other warehouses, after all, remain in business successfully; why not genuine safekeeping places for money? Arthur Nussbaum, *Money in the Law: National and International* (Brooklyn: Foundation Press, 1950), p. 105.

¹¹The economist, Jevons, in a cry from the heart, lamented the existence of the general deposit, since it has “become possible to create a fictitious supply of a commodity, that is, to make people believe that a supply exists which does not exist . . .” On the other hand, special deposits, such as “bills of lading, pawn-tickets, dock-warrants, or certificates which establish ownership to a definite object,” are superior because “they cannot possibly be issued in excess of the good actually deposited, unless by distinct fraud.” He concluded wistfully that “it used to be held as a general rule of law, that a present grant or assignment of goods not in existence is without operation.” William Stanley Jevons, *Money and the Mechanism of Exchange*, 15th ed. (London: Kegan Paul, 1905), pp. 206–12, 221.

The Rothbard Deposit Bank

Assets		Equity & Liabilities	
Gold coin or bullion	\$50,000	Warehouse receipts for gold	\$50,000
Total Assets	\$50,000	Total Liabilities	\$50,000

FIGURE 7.1 — A DEPOSIT BANK

nothing exceptionable has happened. Fifty thousand dollars' worth of gold has simply been deposited in a bank, after which the warehouse receipts circulate from hand to hand or from bank to bank as a surrogate for the gold in question. No fraud has been committed and no inflationary impetus has occurred, because the Rothbard Bank is still backing all of its warehouse receipts by gold or cash in its vaults.

The amount of cash kept in the bank's vaults ready for instant redemption is called its *reserves*. Hence, this form of honest, non-inflationary deposit banking is called "100 percent reserve banking," because the bank keeps all of its receipts backed fully by gold or cash. The fraction to be considered is

$$\frac{\text{Reserves}}{\text{Warehouse Receipts}}$$

and in our example the fraction is

$$\frac{\$50,000}{\$50,000}$$

or 100 percent. Note, too, that regardless of how much gold is deposited in the banks, the total money supply remains precisely the same so long as each bank observes the 100 percent rule. Only the *form* of the money will change, not its total amount or

its significance. Thus, suppose that the total money supply of a country is \$100,000,000 in gold coin and bullion, of which \$70,000,000 is deposited in banks, the warehouse receipts being fully backed by gold and used as a substitute for gold in making monetary exchanges. The total money supply of the country (that is, money actually used in making exchanges) would be:

\$30,000,000 (gold) + \$70,000,000 (warehouse receipts for gold)

The total amount of money would remain the same at \$100,000,000; its form would be changed to mainly warehouse receipts for gold rather than gold itself.

The irresistible temptation now emerges for the goldsmith or other deposit banker to commit fraud and inflation: to engage, in short, in *fractional reserve banking*, where total cash reserves are *lower*, by some fraction, than the warehouse receipts outstanding. It is unlikely that the banker will simply abstract the gold and use it for his own consumption; there is then no likelihood of *ever* getting the money should depositors ask to redeem it, and this act *would* run the risk of being considered embezzlement. Instead, the banker will either lend out the *gold*, or far more likely, will issue fake warehouse receipts for gold and lend them out, eventually getting repaid the principal plus interest. In short, *the deposit banker has suddenly become a loan banker*; the difference is that he is not taking his own savings or borrowing in order to lend to consumers or investors. Instead he is taking someone else's money and lending it out *at the same time* that the depositor thinks his money is still available for him to redeem. Or rather, and even worse, the banker issues fake warehouse receipts and lends them out as if they were real warehouse receipts represented by cash. At the same time, the original depositor thinks that his warehouse receipts are represented by money available at any time he wishes to cash them in. Here we have the system of *fractional reserve banking*, in which more than one warehouse receipt is backed by the same amount of gold or other cash in the bank's vaults.

It should be clear that modern fractional reserve banking is a shell game, a Ponzi scheme, a fraud in which fake warehouse receipts are issued and circulate as equivalent to the cash supposedly represented by the receipts.

Let us see how this works in our T-accounts.

The Rothbard Bank, having had \$50,000 of gold coin deposited in it, now issues \$80,000 of fraudulent warehouse receipts and lends them to Smith, expecting to be repaid the \$80,000 plus interest.

The Rothbard Bank

Assets		Equity & Liabilities	
Gold coin	\$50,000	Warehouse receipts	
IOU from Smith	\$80,000	for gold	\$130,000
Total Assets	\$130,000	Total Liabilities	\$130,000

FIGURE 7.2 — FRACTIONAL RESERVE BANKING

The Rothbard Bank has issued \$80,000 of fake warehouse receipts which it lends to Smith, thus increasing the total money supply from \$50,000 to \$130,000. The money supply has increased by the precise *amount of the credit*—\$80,000—expanded by the fractional reserve bank. One hundred percent reserve banking has been replaced by fractional reserves, the fraction being

$$\frac{\$50,000}{\$130,000}$$

or 5/13.

Thus, fractional reserve banking is at one and the same time fraudulent and inflationary; it generates an increase in the money supply by issuing fake warehouse receipts for money. Money in

circulation has increased by the amount of warehouse receipts issued beyond the supply of gold in the bank.

The form of the money supply in circulation has again shifted, as in the case of 100 percent reserve banking: A greater proportion of warehouse receipts to gold is now in circulation. But something new has now been added: The total amount of money in circulation has now been increased by the new warehouse receipts issued. Gold coin in the amount of \$50,000 formerly in circulation has now been replaced by \$130,000 of warehouse receipts. The lower the fraction of the reserve, the greater the amount of new money issued, pyramiding on top of a given total of reserves.

Where did the money come from? It came—and this is the most important single thing to know about modern banking—it came out of thin air. *Commercial banks—that is, fractional reserve banks—create money out of thin air.* Essentially they do it in the same way as *counterfeiters*. Counterfeiters, too, create money out of thin air by printing something masquerading as money or as a warehouse receipt for money. In this way, they fraudulently extract resources from the public, from the people who have genuinely earned their money. In the same way, fractional reserve banks counterfeit warehouse receipts for money, which then circulate as equivalent to money among the public. There is one exception to the equivalence: The law fails to treat the receipts as counterfeit.

Another way of looking at the essential and inherent unsoundness of fractional reserve banking is to note a crucial rule of sound financial management—one that is observed everywhere *except* in the banking business. Namely, *that the time structure of the firm's assets should be no longer than the time structure of its liabilities*. In short, suppose that a firm has a note of \$1 million due to creditors next January 1, and \$5 million due the following January 1. If it knows what is good for it, it will arrange to have assets of the same amount falling due on these dates or a bit earlier. That is, it will have \$1 million coming due to it before or on January 1, and \$5 million by the year following. Its time structure of assets is no longer, and preferably a bit shorter, than its liabilities

coming due. But deposit banks do not and *cannot* observe this rule. On the contrary, its liabilities—its warehouse receipts—are due instantly, on demand, while its outstanding loans to debtors are inevitably available only after some time period, short or long as the case may be. A bank's assets are always "longer" than its liabilities, which are instantaneous. Put another way, *a bank is always inherently bankrupt*, and would actually become so if its depositors all woke up to the fact that the money they believe to be available on demand is actually not there.¹²

One attempted justification of fractional reserve banking, often employed by the late Professor Walter E. Spahr, maintains that the banker operates somewhat like a bridge builder. The builder of a bridge estimates approximately how many people will be using it from day to day; he doesn't attempt the absurd task of building a bridge big enough to accommodate every resident of the area should he or she wish to travel on the bridge at the same time. But if the bridge builder may act on estimates of the small fraction of citizens who will use the bridge at any one time, why may not a banker likewise estimate what percentage of his deposits will be redeemed at any one time, and keep no more than the required fraction? The problem with this analogy is that citizens in no sense have a legal claim to be able to cross the bridge at any given time. But holders of warehouse receipts to money emphatically *do have* such a claim, even in modern banking law, to their own property any time they choose to redeem it. But the legal claims issued by the bank must then be fraudulent, since the bank could not possibly meet them all.¹³

¹²Cf. Elgin Groseclose, *Money and Man*, pp. 178–79.

¹³See Murray N. Rothbard, *The Case for a 100 Percent Gold Dollar* (Washington, D.C.: Libertarian Review Press, November 1974), p. 25. Mises trenchantly distinguishes between a "credit transaction," where a present good is exchanged for a future good (or IOU due in the future), and a *claim transaction*, such as a warehouse receipt, where the depositor or claimant does not give up any of the present good (e.g., wheat, or money). On the contrary, he retains his claim to the deposited good, since he can redeem it at any time. As Mises states:

It should be clear that for the purpose of analyzing fractional reserve banking, it doesn't make any difference what is considered money or cash in the society, whether it be gold, tobacco, or even government fiat paper money. The technique of pyramiding by the banks remains the same. Thus, suppose that now gold has been outlawed, and cash or legal tender money consists of *dollars* printed by the central government. The process of pyramiding remains the same, except that the base of the pyramid is paper dollars instead of gold coin.¹⁴

Our Rothbard Bank which receives \$50,000 of government paper money on deposit, then proceeds to pyramid \$80,000 on top of it by issuing fake warehouse receipts.

The Rothbard Bank

Assets		Equity & Liabilities	
Gold coin	\$50,000	Warehouse receipts	
IOU from Smith	\$80,000	to cash	\$130,000
Total Assets	\$130,000	Total Liabilities	\$130,000

FIGURE 7.3 — FRACTIONAL RESERVE BANKING (PAPER)

A depositor of a sum of money who acquires in exchange for it a claim convertible into money at any time which will perform exactly the same service for him as the sum it refers to has exchanged no present good for a future good. The claim that he has acquired by his deposit is also a present good for him. The depositing of money in no way means that he has renounced immediate disposal over the utility it commands.

Ludwig von Mises, *The Theory of Money and Credit*, 2nd ed. (New Haven: Yale University Press, 1953), p. 268.

¹⁴As we shall see later, while the *pyramiding process* remains the same, the opportunity for inflating the base is much greater under fiat paper than with gold.

Just as in the gold case, the total money supply has increased from \$50,000 to \$130,000, consisting precisely in the issue of new warehouse receipts, and in the credit expanded by the fractional reserve bank.

Just as in the case of outright counterfeiting, the new money—this time in the form of new warehouse receipts—does not shower upon everyone alike. The new money is injected *at some particular point* in the economic system—in this case, the Rothbard Bank issues it and it is immediately loaned to Smith—and the new money then ripples out into the economy. Smith, let us say, uses the \$80,000 of new money to buy more equipment, the equipment manufacturer buys raw materials and pays more for labor, and so on. As the new money pours into the system and ripples outward, demand curves for particular goods or services are increased along the way, and prices are increased as well. The more extensive the spread of bank credit, and the more new money is pumped out, the greater will be its effect in raising prices. Once again, the early receivers from the new money benefit at the expense of the late receivers—and still more, of those who never receive the new money at all. The earliest receivers—the bank and Smith—benefit most, and, like a hidden tax or tribute, the late receivers are fraudulently despoiled of their rightful resources.

Thus, fractional reserve banking, like government fiat paper or technical counterfeiting, is inflationary, and aids some at the expense of others. But there are even more problems here. Because unlike government paper and unlike counterfeiting (unless the counterfeit is detected), the bank credit is subject to contraction as well as expansion. In the case of bank credit, what comes up, can later come down, and generally does. The expansion of bank credit makes the banks shaky and leaves them open, in various ways, to a contraction of their credit.

Thus, let us consider the Rothbard Bank again. Suppose that the loan to Smith of \$80,000 was for a two-year period. At the end of the two years, Smith is supposed to return the \$80,000 plus interest. But when Smith pays the \$80,000 (forgetting about

the interest payment to keep things simple), he will very likely pay in Rothbard Bank warehouse receipts, which are then canceled. The repayment of the \$80,000 loan means that \$80,000 in fake warehouse receipts has been canceled, and the money supply has now contracted back to the original \$50,000. After the repayment, the balance sheet of the Rothbard Bank will be as follows:

<i>The Rothbard Bank</i>			
Assets		Equity & Liabilities	
Cash	\$50,000	Warehouse receipts to cash	\$50,000
Total Assets	\$50,000	Total Liabilities	\$50,000

FIGURE 7.4 — REPAYMENT OF BANK LOANS

We are back to the pre-expansion figures of our original example (Figure 7.1).

But if the money supply contracts, this means that there is deflationary pressure on prices, and prices will contract, in a similar kind of ripple effect as in the preceding expansion. Ordinarily, of course, the Rothbard Bank, or any other fractional reserve bank, will not passively sit back and see its loans and credit contract. Why should it, when the bank makes its money by inflationary lending? But, the important point is that fractional reserve banks are sitting ducks, and are always subject to contraction. When the banks' state of inherent bankruptcy is discovered, for example, people will tend to cash in their deposits, and the contractionary, deflationary pressure could be severe. If banks have to contract suddenly, they will put pressure on their borrowers, try to call in or will refuse to renew their loans, and the deflationary

pressure will bring about a recession—the successor to the inflationary boom.

Note the contrast between fractional reserve banking and the pure gold coin standard. Under the pure gold standard, there is virtually no way that the money supply can actually *decline*, since gold is a highly durable commodity. Nor will it be likely that government fiat paper will decline in circulation; the only rare example would be a budget surplus where the government burned the paper money returning to it in taxes. But fractional reserve bank credit expansion is always shaky, for the more extensive its inflationary creation of new money, the more likely it will be to suffer contraction and subsequent deflation. We already see here the outlines of the basic model of the famous and seemingly mysterious business cycle, which has plagued the Western world since the middle or late eighteenth century. For every business cycle is marked, and even ignited, by inflationary expansions of bank credit. The basic model of the business cycle then becomes evident: bank credit expansion raises prices and causes a seeming boom situation, but a boom based on a hidden fraudulent tax on the late receivers of money. The greater the inflation, the more the banks will be sitting ducks, and the more likely will there be a subsequent credit contraction touching off liquidation of credit and investments, bankruptcies, and deflationary price declines. This is only a crude outline of the business cycle, but its relevance to the modern world of the business cycle should already be evident.

Establishing oneself as a fractional reserve bank, however, is not as easy as it seems, despite the law unfortunately looking the other way at systemic fraud. For the Rothbard Bank, or any other bank, to have its warehouse receipts functioning in lieu of gold or government paper requires a long initial buildup of trust on the part of the public. The Rothbard Bank must first build up a reputation over the decades as a bank of safety, probity, and honesty, and as always ready and able to redeem its liabilities on demand. This cannot be achieved overnight.

4. BANK NOTES AND DEPOSITS

Through the centuries, there have been two basic forms of money warehouse receipts. The first, the most obvious, is the written receipt, a piece of paper on which the deposit bank promises to pay to the bearer a certain amount of cash in gold or silver (or in government paper money). This written form of warehouse receipt is called the *bank note*. Thus, in the United States before the Civil War, hundreds if not thousands of banks issued their own notes, some in response to gold deposited, others in the course of extending fractional reserve loans. At any rate, if someone comes into the possession (either by depositing gold or by selling a product in exchange) of, say, a \$100 note from the Bank of New Haven, it will function as part of the money supply so long as people accept the \$100 note as a substitute, a surrogate, for the gold. If someone uses the \$100 note of the Bank of New Haven to buy a product sold by another person who is a customer of the Bank of Hartford, the latter will go to his bank and exchange the \$100 New Haven note for a similar note from the Bank of Hartford.

The bank note has always been the basic form of warehouse receipt used by the mass of the public. Later, however, there emerged another form of warehouse receipt used by large merchants and other sophisticated depositors. Instead of a tangible receipt, the bank simply opened a deposit account on its books. Thus, if Jones deposited \$10,000 in a bank, he received, if he wished, not tangible bank notes, but an open book account or deposit account for \$10,000 on the bank's books. The bank's demand debt to Jones was not in the form of a piece of paper but of an intangible book account which could be redeemed at any time in cash. Confusingly, these open book accounts came to be called *demand deposits*, even though the tangible bank note was just as much a demand deposit from an economic or a legal point of view. When used in exchange, instead of being transferred physically as in the case of a bank note, the depositor, Jones, would write out an order, directing the bank to transfer his book

account to, say, Brown. Thus, suppose that Jones has a deposit account of \$10,000 at the Rothbard Bank.

Suppose now that Jones buys a hi-fi set from Brown for \$3,000. Jones writes out an order to the bank, directing it to transfer \$3,000 from his open book account to that of Brown. The order will appear somewhat as follows:

Rothbard Bank
 Pay to the order of John Brown \$3,000
 Three thousand and 00/000
 (signed)
 Robert Jones

This written instrument is, of course, called a *check*. Note that the check *itself* is not functioning as a money surrogate here. The check is simply a written order transferring the demand deposit from one person to another. The demand deposit, not the check, functions as money, for the former is a warehouse receipt (albeit unwritten) for money or cash.

The Rothbard Bank's balance sheet is now as follows:

<i>The Rothbard Bank</i>			
Assets		Equity & Liabilities	
Gold	\$10,000	Demand deposits	
		to Jones	\$7,000
		to Brown	\$3,000
Total Assets	\$10,000	Total Liabilities	\$10,000

FIGURE 7.5 — TRANSFERRING DEMAND DEPOSITS

Note that from this purchase of a hi-fi set, nothing has changed in the total money supply in the country. The bank was and still is pursuing a 100 percent reserve policy; all of its demand liabilities

are still covered or backed 100 percent by cash in its vaults. There is no fraud and no inflation.

Economically, then, the demand deposit and the tangible bank note are simply different technological forms of the same thing: a demand receipt for cash at the money warehouse. Their economic consequences are the same and there is no reason for the legal system to treat them differently. Each form will tend to have its own technological advantages and disadvantages on the market. The bank note is simpler and more tangible, and doesn't require quite the same degree of sophistication or trust by the holders of the receipt. It also involves less work for the bank, since it doesn't have to change the names on its books; all it needs to know is that a certain quantity of bank notes is out in circulation. If Jones buys a hi-fi set from Brown, the bank note changes hands without anyone having to report the change at the bank, since the bank is liable to the note-holder in any case. For small transactions—purchase of a newspaper or ham sandwich—it is difficult to visualize having to write out a check in payment. On the other hand, demand deposits have the advantage of allowing one to write out checks for exact amounts. If, for example, the hi-fi set costs some nonrounded amount, such as \$3,168.57, it may well be easier to simply write out the check than trying to find notes and coins for the exact amount—since notes will generally be in fixed denominations (\$1, \$5, \$10, etc.).¹⁵ Also, it will often be more convenient to use demand deposits for large transactions, when amassing cash can be cumbersome and inconvenient. Moreover, there is far greater danger of loss from theft or accident when carrying cash than when having a certain known amount on a bank's books.

¹⁵Bank notes, however, were made more flexible in seventeenth-century England by the banks allowing *part* payment of a note, with the payment deducted from the original face value of the note. Holden, *Negotiable Instruments*, p. 91n.

All of these factors will tend, on the free market, to limit the use of bank deposits to large users and for large transactions.¹⁶ As late as World War I, the general public in the Western world rarely used bank deposits. Most transactions were effected in cash, and workers received cash rather than bank checks for wages and salaries. It was only after World War II, under the impetus of decades of special support and privilege by government, that checking accounts became nearly universal.

A bank can issue fraudulent and inflationary warehouse receipts just as easily in the form of open book deposits as it can in bank notes. To return to our earlier example, the Rothbard Bank, instead of printing fraudulent, uncovered bank notes worth \$80,000 and lending them to Smith, can simply open up a new or larger book account for Smith, and credit him with \$80,000, thereby, at the stroke of a pen and as if by magic, increasing the money supply in the country by \$80,000.

In the real world, as fractional reserve banking was allowed to develop, the rigid separation between deposit banking and loan banking was no longer maintained in what came to be known as *commercial banks*.¹⁷ The bank accepted deposits, loaned out its

¹⁶ . . . banking in general only became important with the development of the issue of notes. People would deposit coin and bullion with a bank more readily when they received something in exchange such as a banknote, originally in the form of a mere receipt, which could be passed from hand-to-hand. And it was only after the bankers had won the public over to confidence in the banks by circulating their notes, that the public was persuaded to leave large sums on deposit on the security of a mere book-entry.

Vera C. Smith, *The Rationale of Central Banking* (London: P.S. King & Son, 1936), p. 6.

¹⁷The later institution of the “investment bank,” in contrast, lends out saved or borrowed funds, generally in the underwriting of industrial or government securities. In contrast to the commercial bank, whose deposit liabilities exchange as equivalent to money and hence add to the money supply, the liabilities of the investment bank are simply debts which are not “monetized” by being a demand claim on money.

equity and the money it borrowed, and also created notes or deposits out of thin air which it loaned out to its own borrowers. On the balance sheet, all these items and activities were jumbled together. Part of a bank's activity was the legitimate and productive lending of saved or borrowed funds; but most of it was the fraudulent and inflationary creation of a fraudulent warehouse receipt, and hence a money surrogate out of thin air, to be loaned out at interest.

Let us take a hypothetical mixed bank, and see how its balance sheet might look, so that we can analyze the various items.

Jones Bank

Assets		Equity & Liabilities	
IOUs from		Demand Liabilities:	
borrowers	\$1,700,000	Notes	\$1,000,000
Cash	\$300,000	Deposits	\$800,000
		Total	\$1,800,000
		Equity	\$200,000
Total Assets	\$2,000,000	Total Liabilities	\$2,000,000

FIGURE 7.6 — MIXED LOAN AND DEPOSIT BANK

Our hypothetical Jones Bank has a stockholders' equity of \$200,000, warehouse receipts of \$1.8 million distributed as \$1 million of bank notes and \$800,000 of demand deposits, cash in the vault of \$300,000, and IOUs outstanding from borrowers of \$1.7 million. Total assets, and total equity and liabilities, each equal \$2 million.

We are now equipped to analyze the balance sheet of the bank from the point of view of economic and monetary importance. The crucial point is that the Jones Bank has demand liabilities, instantly payable on presentation of the note or deposit, totaling

\$1.8 million, whereas cash in the vault ready to meet these obligations is only \$300,000.¹⁸ The Jones Bank is engaging in fractional reserve banking, with the fraction being

$$\frac{\$300,000}{\$1,800,000}$$

or 1/6. Or, looking at it another way, we can say that the invested stockholder equity of \$200,000 is invested in loans, while the other \$1.5 million of assets have been loaned out by the creation of fraudulent warehouse receipts for money.

The Jones Bank could increase its equity by a certain amount, or borrow money by issuing bonds, and then invest them in extra loans, but these legitimate loan operations would not affect the 1/6 fraction, or the amount of fraudulent warehouse receipts outstanding. Suppose, for example, that stockholders invest another \$500,000 in the Jones Bank, and that this cash is then loaned to various borrowers. The balance sheet of the Jones Bank would now appear as shown in Figure 7.7.

Thus, while the Jones Bank has extended its credit, and its new extension of \$500,000 of assets and liabilities is legitimate, productive and noninflationary, its inflationary issue of \$1,500,000 continues in place, as does its fractional reserve of 1/6.

A requirement that banks act as any other warehouse, and that they keep their demand liabilities fully covered, that is, that they engage only in 100 percent banking, would quickly and completely put an end to the fraud as well as the inflationary impetus of modern banking. Banks could no longer add to the money supply, since they would no longer be engaged in what is tantamount

¹⁸We should note, however, that if they wanted to, the holders of \$800,000 of the bank's demand deposits could cash them in for the notes of the Jones Bank, as well as for gold or government paper money. In fact, the notes and deposits of the Jones Bank are interchangeable for each other, one for one: deposits could, if the owner wished, be exchanged for newly-printed notes, while notes could be handed in and exchanged for newly-credited deposits.

Jones Bank

Assets		Equity & Liabilities	
IOUs from borrowers	\$2,200,000	Demand Liabilities:	
Cash	\$300,000	Notes	\$1,000,000
		Deposits	\$800,000
		Total	\$1,800,000
		Equity	\$700,000
Total Assets	\$2,500,000	Total Liabilities	\$2,500,000

FIGURE 7.7 — FRACTIONAL RESERVE IN A MIXED BANK

to counterfeiting. But suppose that we don't have a legal requirement for 100 percent banking. How inflationary would be a system of free and unrestricted banking, with no government intervention? Is it true, as is generally believed, that a system of free banking would lead to an orgy of unrestricted money creation and inflation?